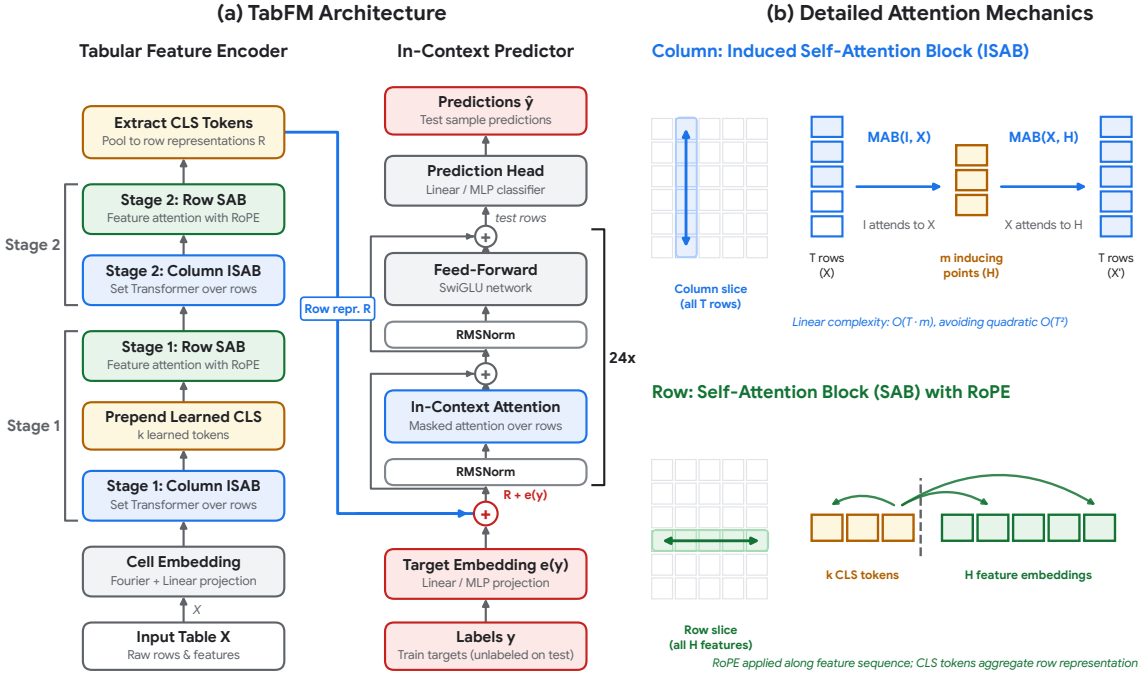

(a) TabFM Architecture
Tabular Feature Encoder
In-Context Predictor
Extract CLS Tokens
Pool to row representations R
Stage 2
Stage 2: Row SAB
Feature attention with RoPE
Stage 2: Column ISAB
Set Transformer over rows
Stage 1
Stage 1: Row SAB
Feature attention with RoPE
Prepend Learned CLS
k learned tokens
Stage 1: Column ISAB
Set Transformer over rows
Cell Embedding
Fourier + Linear projection
X
Input Table X
Raw rows & features
Row repr. R
Predictions ŷ
Test sample predictions
Prediction Head
Linear / MLP classifier
test rows
Feed-Forward
SwiGLU network
RMSNorm
In-Context Attention
Masked attention over rows
RMSNorm
24x
R + e(y)
Target Embedding e(y)
Linear / MLP projection
Labels y
Train targets (unlabeled on test)
(b) Detailed Attention Mechanics
Column: Induced Self-Attention Block (ISAB)
MAB(I, X)
MAB(X, H)
I attends to X
X attends to H
T rows
(X)
m inducing
points (H)
T rows
(X')
Column slice
(all T rows)
Linear complexity: O(T · m), avoiding quadratic O(T²)
Row: Self-Attention Block (SAB) with RoPE
k CLS tokens
H feature embeddings
Row slice
(all H features)
RoPE applied along feature sequence; CLS tokens aggregate row representation R